Experimentally Determining the Resistivity of Constantan Wire

A Laboratory Report and Practical Writeup

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1. Abstract

Resistivity, also known as specific electrical resistance, is a bulk property that quantifies a material's ability to restrict a flow of electrical current. We present and analyse a method for experimentally determining resistivity of a wire by varying the cross-sectional area of the wire, and report a value of $\rho = (4.83 \pm 1.32) \times 10^{-7}\ \Omega\text{m}$ for the resistivity of constantan, a metal alloy conductor of 55% copper and 45% nickel, also known as Eureka. This interval encloses the true value[11] of $\rho = 4.9 \times 10^{-7}\ \Omega\text{m}$ and lies closely to the mean value of our interval. We also discuss uncertainties due to the apparatus used, provide a statistical treatment of the data collected, concluding our results were accurate and enclose the true value at the 99.9% significance level. We also show that Joule heating (the increase in resistance due to current flow and temperature rise) played no significant role in modifying our results. We end the analysis with a comparison between two methods of finding ρ, which returned no difference in accuracy but a slight favour in precision to our method, and an explanation of the differences in reliability between the methods.

Keywords: Resistivity, Constantan, Wire, Experiment

2. Introduction

An electric current is produced when a potential difference is applied between two points on a conductor. The simplest case is when the potential difference is created by a battery or other power source.

The resistance due to an applied potential difference V inducing a current I in a conductor is given by a re-arrangement of Ohm's law, namely $R = V/I$. (Equation 1). This potential difference creates an electromotive force which acts on charge carriers in the wire, causing them to move. A perfect conductor would allow these charge carriers to move without restriction, such that when the source of p.d. is removed, the charge carriers continue to move around the circuit. In reality only a very small class of materials (known as *superconductors*) permit this, while the vast majority of conductors inhibit current flow due to collisions between the electrons as well as with the metal ion lattice. This causes the charge carriers to lose energy and hence a decrease in current. The ratio of the applied potential difference to the current in the conductor is defined as the *resistance* of the conductor.

Since resistance is dependent on collisions between charge carriers, it is affected by the amount of conductor material present, as well as the state of the system. To make the concept of resistance more fundamental and applicable to a wider range of physics and engineering problems, a quantity known as *resistivity* was derived to eliminate the dependence on the quantity of the conductor.

In practical engineering contexts, quantification of resistivity is indicative of the energy loss (i.e. the power dissipation) of an incoming electric current. Selecting materials for transmission cables, for example, is crucial in ensuring the maximum efficiency in such large-scale power transfer projects, with £965 million being spent in 2018 by the National Grid to maintain and improve power transmission in the UK.[5]

3. Theory

Resistivity is defined quantitatively by

$$\rho = \frac{AR}{L} \quad \text{(Equation 2)}$$

where A is the cross-sectional area of the material, R is the electrical resistance and L is the length of material. It is a quantity assigned to materials describing their ability to restrict the flow of current through them. It is the bulk analogue of resistance, being constant for any material at a given temperature. Through dimensional analysis it can be deduced that the units of resistivity are Ωm. For instance, rubber, a prime example of an insulator, has a very high resistivity on the order of 10^{14} Ωm, while conductors such as copper and most metals have much lower resistivities, generally on the order of 10^{-8} Ωm. Superconductors, by definition, have exactly zero resistivity below a certain critical temperature due to the formation of a new quantum state in which electrons no longer interact with the atoms in the material[10].

There are three main factors affecting resistivity. Electrical resistance arises due to the collision of conducting electrons with ions as well as with each other. In metals, conduction occurs due to the potential-induced movement of delocalised electrons surrounding a lattice of positive ions. When the electrons must travel a long distance, there is more opportunity for such collisions to occur and hence we expect a higher value of resistivity with a higher length. Conversely, for a wire of larger cross-sectional area, there are more free charge carriers per unit length allowing a higher flow of charge.

The third factor affecting material resistivity is temperature. A higher temperature means the metal ions vibrate with a higher energy and hence a higher amplitude of oscillation, which restricts electron mobility due to more frequent collisions. This is slightly problematic since the energy lost due to electrical resistance is transferred directly to the thermal energy store of the wire, increasing temperature and further increasing resistance, a phenomenon known as Joule heating. The energy transferred per second (power loss) is given by

$$P = \frac{E}{t} = I^2 R \quad \text{(Equation 3)}$$

Assuming direct energy transfer, this energy increases the temperature of the metal according to $E = mc\Delta T$ (Equation 4), where c is the *specific heat capacity* of the metal. Using the definition of density ρ, we substitute $m = \rho AL$ and equating Equations 3 and 4 yields

$$\Delta T = \frac{I^2 Rt}{c\rho AL} \text{ (Equation 5)}$$

which can then be used to approximate the resistance at the original temperature through[2]

$$R_0 = \frac{R_T}{1 + \alpha \Delta T} \text{ (Equation 6)}$$

where α is the *temperature coefficient of resistance.* In reality, the thermal energy is not transferred solely to the metal, but also to the surrounding air, which would require more in-depth analysis of the relevant partial differential equations, namely the *heat equation*: [7][12]

$$\frac{\partial T}{\partial t} = \kappa \cdot \frac{\partial^2 T}{\partial x^2} + \frac{E}{c\rho} \text{ (Equation 7)}$$

where E is the heat energy produced due to Joule heating as described by Equation 3 and κ is the thermal diffusivity of the material. In this case there are two interdependent variables; temperature and resistance, which are modelled as functions of time and a constant potential. The resistance term is also constrained by Maxwell's equations due to its dependence on the current flow.

The relevant physical properties of the constantan we used are c = 390 J kg^{-1} K^{-1}, ρ = 8900 kg m^{-3}, α = 3 × 10^{-5} K^{-1} and κ = 6.6 × 10^{-6} m^2 s^{-1} [3][11]. The equations discussed above could have been used to attempt a correction for Joule heating within our experiment, however the effect of such a correction is negligible due to the very small temperature coefficient of constantan used.

4. Experiment Setup and Procedure

4.1. Apparatus and Safety Precautions

When working with electricity, precautions were taken to avoid injury[1]. The hazards identified are outlined in Table 1:

Table 1

Source of danger	Hazard	Risk	Precaution
Power supply	Heavy equipment	Low	Carry the power supply carefully to the workspace. Do not move the power supply once it is on the workspace and ensure its stability if necessary.
Wires	Electrocution	Medium	Do not touch the wire when the power supply is turned on. Take voltage readings by holding the insulating grips on the jockeys and not the conductive parts.
	Heat burns	Medium	Do not leave the power supply on for extended periods of time. Pick up the wire using the wooden ruler attached to it and not holding the wire itself.
	Electrical fires	Very low	Avoid using currents above 2 A.
	Sharp ends	Low	Take care when carrying the wire between places. Do not point the wooden ruler at other people.
	Trip hazard	Very low	Ensure the wire remains on the workspace at all times. Be observant when carrying the wire.
Jockeys and Crocodile Clips	Sharp	Very low	Grip the jockeys firmly and point them away from oneself when carrying them.

The apparatus we used is outlined in *Table 2*:

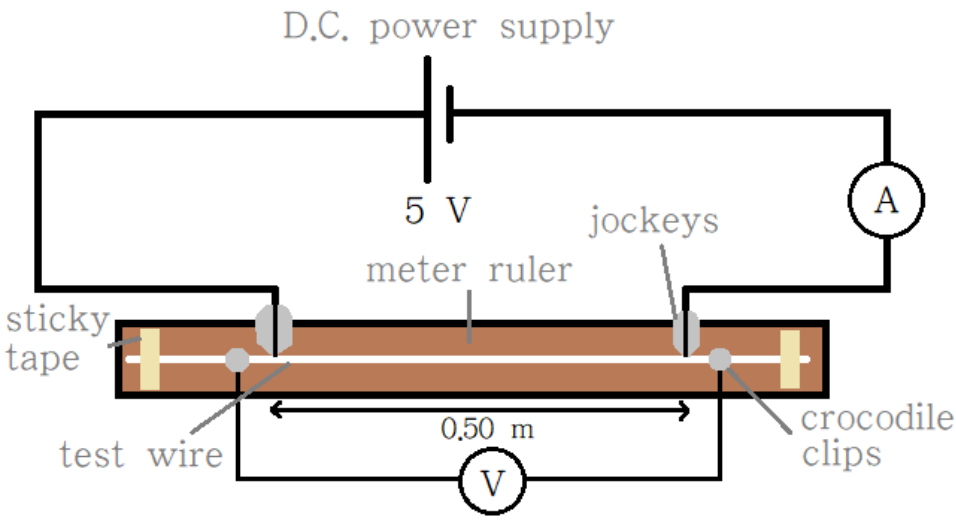
Table 2

Apparatus	Purpose
DC power supply	To provide a potential difference in the circuit
7 samples of constantan wire of different cross-sectional area, labelled A-G	To provide various samples to test the resistance across
Digital micrometer	To accurately determine the diameter of the wire
Ammeter	To determine the current flowing in the test wire
Voltmeter	To determine the voltage drop across the test wire
Insulating wires	To complete the circuit, connecting the test wire to the power supply safely
Wooden meter ruler	To ensure a constant length of wire is used in each test
Jockeys	To provide a more efficient electrical connection between the test wire and the insulating wires
Sandpaper	To remove contaminants from the jockeys, improving their performance

4.2. Setup

A simple circuit was set up as shown in *Figure 1*:

Figure 1



The seven test wires, labelled A through G, were measured for their diameter using a digital micrometer, which was calibrated before each measurement. The measurement was taken at three points along the wire – at each end and in the middle, and the mean value and uncertainty was recorded. A table of cross-sectional areas, calculated using $A = \pi r^2/4$ (Equation 8), was constructed.

The jockeys were sharpened and cleaned using rough sandpaper before each test in order to minimise resistance at the terminals.

The test wires were straightened and held down using sticky tape before each test to ensure that the correct fixed length of 0.50 m was being used throughout.

4.3. Method

Once the apparatus had been prepared and setup, the power supply was turned on and set to a constant voltage of 5 V. For each of the test wires, we:

1. Held the jockeys to the wire between the marked points of 0.25 m and 0.75 m on the ruler

2. Read the voltmeter and ammeter and noted their fluctuations

3. Turned off the power supply

4. Calculated the resistance using Equation 1.

4.4. Graphing and Obtained Results

Once values of *R* and *A* were obtained for each test wire, we rearranged Equation 2 to the form of a straight-line, i.e.

$$\rho = \frac{AR}{L} \rightarrow R = \frac{L}{A}\rho \leftrightarrow R = \frac{0.5}{A} \cdot \rho \leftrightarrow y = mx$$

where L = 0.5 m was specified as the control variable in our experiment, *A* was the independent variable and *R* was the dependent variable.

From our results, the graph we obtained in Excel for plotting *R* against 0.5/*A* is shown in *Figure 2*. Absolute uncertainties in both variables are shown by error bars and a line of best fit was constructed as shown in red, and a line of worst fit was constructed in dotted orange.

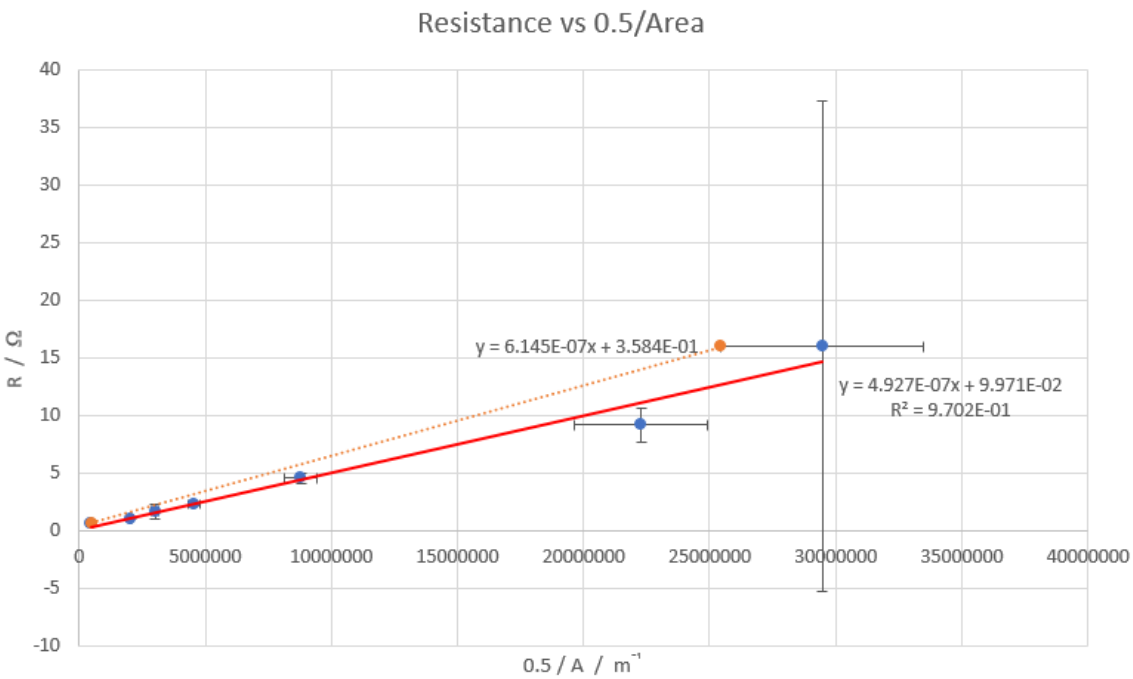


Figure 2

The gradient of the line of best fit is 4.83×10^{-7}, which represents our value for the mean resistivity in Ωm. A worst-fit line was defined as the line connecting the lower bound on the maximum value of *R* to the upper bound of the minimum value of *R*. The gradient of this line was 6.15×10^{-7}. The difference between these gradients was 1.32×10^{-7}, which left us with a final result of $\rho = (4.83 \pm 1.32) \times 10^{-7}$ Ωm.

5. Analysis

We present our analysis of the results obtained, conducted after the experiment.

5.1. Uncertainties

Sources of uncertainty in our experiment come from the apparatus used, as well as slight inconsistencies between measurements due to human error.

5.1.1. Identification and Quantification of Uncertainty

The apparatus we used has various orders of magnitude of resolution. *Table 3* shows the absolute uncertainties of the apparatus we used, as well as the variation in readings where relevant.

Table 3

Apparatus	**Absolute Uncertainty due to Resolution**	**Absolute Uncertainty due to Random Variation**	**Quantity Affected**	**Total Percentage uncertainty range (throughout dataset)**
Ruler	± 0.001 m	N/A (constant)	Length	0.2% (negligible)
Micrometer	± 0.000 001 m	± 0.000 010 m	Diameter, Area	0.87% - 6.8% (diameter) 1.74% - 13.6% (area)
Voltmeter	± 0.01 V	± 0.05 V	Voltage	1.6% - 23.8%
Ammeter	± 0.01 A	± 0.05 V	Current	2.7% - 125%
Power Supply	N/A	Indeterminate	Voltage, Current	Indeterminate (assumed negligible; taken as zero)

5.1.2. Consequences of Uncertainty

There were two sources of uncertainty for the micrometer, voltmeter and ammeter. The resolution of the instrument was usually small compared to the readings being done, although this was not always the case. There was also a second source of uncertainty in the random variation of the measurements throughout the experiment. The diameter of the wire was not constant but varied slightly. Only the mean value was included in the subsequent calculations. The readings of V and I fluctuated significantly, up to ± 0.05 units, causing drastic uncertainty in these measurements: in particular, an impressive uncertainty of 125% in one reading of I arose due to measuring a current less than the variable range of the ammeter.

5.1.3. Statistical Treatment

The largest result in *Figure 2* has unfeasibly large error bars, to the extent that they reach values of negative resistance which is clearly nonsensical. Therefore, we have chosen to exclude this data point in the subsequent analysis.

The line of best fit for the data had an R-value (correlation coefficient) of 0.985, indicating a very strong positive correlation. Application of Spearman's Rank Test[8] returns perfect correlation at the 99.9% significance level.

Taking the following hypotheses as standard,

H_0: The true value of ρ does not lie within our specified range.

H_1: The true value of ρ does in fact lie within our specified range.

We defined the c.r.v X to be $X / 10^{-7} = R / (0.5 / A)$. Each observation X_i generated the values shown in *Table 4*.

Table 4

i	1	2	3	4	5	6
$X_i / 10^{-7}$	4.117	5.221	5.066	5.287	5.590	5.431

Analysis of the data in *Table 4* returned a mean and variance such that X is distributed according to $X \sim N(\mu = 5.11, \sigma^2 = 0.227)$, assuming values are distributed normally about our best fit line.

From the inverse normal distribution[6], the true value of 4.9 returned a z-score[9] of z = 0.44, which is well within the critical value of 3.09 required at the 99.9% confidence level[4]. As a result, we rejected the null hypothesis with 0.1% probability of Type I error.

Therefore, there is sufficient evidence to support our conclusion that the true value does in fact lie within our range, indicating that all uncertainties have been quantified and accounted for correctly.

5.2. Errors

5.2.1. Identifying Potential Sources of Error

We identified two sources of possible error within our experiment.

Firstly, due to the fact that during the experiment we had to hold the jockeys to the wire for an extended period of time, it is likely that the wire would have been moved slightly between readings, changing its effective length between which we measured the potential difference.

Secondly, the persistent current flow in the wire would have increased the resistance slightly than what we would expect at lower currents, possibly giving higher values for *R* than intended.

5.2.2. Attempted Corrections Accounting For Errors

The first error of variable wire length is very difficult to quantify as the non-linear shape of the wire after being moved means its precise length is unfeasible to measure simply. We instead focussed on a potential correction for Joule heating, and we were able to show that Joule heating would have made insignificant changes to our results considering external uncertainties.

To demonstrate this, we selected the reading with the highest current of I = 1.85 A, which was wire D in our test samples. From Equation 3 we find a total power dissipation of

$$P = I^2 R = 1.85^2 \times 1.08 = 3.70\ W.$$

From Equation 5 we estimated the temperature rise in the wire due to Joule heating, using all relevant physical parameters, to be

$$\Delta T = \frac{Pt}{c\rho AL} = \frac{3.7 \times 5}{390 \times 8900 \times 2.45 \times 10^{-7} \times 0.5} = +43.5\ ^{\circ}\mathrm{C},$$

where we have used t = 5 seconds as all our measurements were made within this time frame. From Equation 6 we then estimated the resistance of the wire at room temperature, taking R_T to be the obtained value of R at the elevated temperature to be

$$R_0 = \frac{R_T}{1 + \alpha \Delta T} = \frac{1.08}{1 + (3 \times 10^{-5} \times 43.5)} = 1.078\ \Omega,$$

which, to 3 significant figures, is a negligible change to our measured value of 1.081 Ω. Further analysis using Equation 7 is unnecessary as accounting for thermal diffusion would only decrease the temperature gradient, making the resistance increase even less relevant. We can therefore safely conclude that in all cases, the low temperature coefficient of constantan ensured that Joule heating played no significant role in modifying our results.

5.3. Results

5.3.1. Accuracy and Repeatability

We have shown that our results are likely accurate as our uncertainty range encloses the true value. Our experiment is also repeatable as the apparatus used is standard in many laboratories and the procedure is simple with few systematic errors.

5.3.2. Reproducibility and Feasibility for Improvement

The results we obtained are reproducible, as demonstrated by comparing our results to other groups. The uncertainty ranges overlapped in all cases, signifying a collectively valid result lying very close to the true value.

This experiment could be improved by using higher currents to minimise the uncertainty of V and I, as these were the variables which propagated the most uncertainty in our results. We have shown that the effect of heating due to high currents is negligible and need not be considered even at higher currents. However, there may be the concern of safety when using higher currents as the risks of electrocution and burns become more immanent with increased energy in the circuit.

6. Comparison with Another Method

We chose to find the resistivity by varying the cross-sectional area while keeping the length constant. In fact there was no need to choose this combination of variables and an alternative method was to vary the length while keeping the cross-sectional area constant.

This method was conducted by another group, who obtained a value of $(4.57 \pm 1.41) \times 10^{-7}$ Ωm.

6.1. Accuracy

The interval obtained by this alternative method encloses the true value, suggesting that there is no inherent difference in the accuracy of the methods.

6.2. Uncertainty and Precision

Despite both methods having large uncertainties, our method is slightly more precise with a percentage uncertainty of 27% compared to the alternative method's 31%. The larger uncertainty may be due to larger fluctuations in the voltmeter and ammeter readings, or lower values of V and I being measured. It is unlikely that the uncertainty came from L, as we showed that this had a negligible influence (less than 0.2%) on overall uncertainty.

6.3. Errors

The alternative method is subject to the same errors as our own method, although there was possibly a greater magnitude of the error in the alternative method, since there was more wire being used and therefore more opportunity for bending and variation in length.

7. Conclusions

From the analysis discussed in this report, we conclude that this experiment has been a success. This has been justified by a statistical evaluation as well as quantification of the uncertainties encountered, supplemented by a demonstration that the potential source of error in Joule heating is negligible. The repeatability of the experiment and reproducibility of the results leads us to conclude that our results are as valid as possible with the given apparatus and procedure.

8. Acknowledgements

The following publication on *cisplatin* was found during independent research to be a useful source of guidance for writing this report, including the layout, structure and written style:

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4146684/#S36title

The content of the above paper is not relevant to the topic discussed in this report and was used solely as layout guidance.

The following document from the University of Sheffield was also used to assist content in this report:

https://www.sheffield.ac.uk/polopoly_fs/1.825931!/file/Guidance_for_Writing_Lab_Reports.pdf

I would also like to thank Bien Salom for showing a simplistic but professional format for the title page of this report.

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